

Dania Orfali

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EXPERIENCE

Lawrence Berkeley National Lab (LBNL)

Research Doctoral Fellow

Jan 2026 – Present

California, USA

- Designed and fabricated a novel gas-Membrane Electrode Assembly (MEA) cell compatible with operando X-ray spectroscopy
- Conducted synchrotron X-ray absorption spectroscopy (XAS) fluorescence experiments
- Performed and analysed XANES and EXAFS measurements to resolve catalyst oxidation state and coordination environment under millisecond scale pulsing electrochemistry, with inline product characterization (GC, MS & NMR)
- Sputtered Cu catalyst films to fabricate gas diffusion electrodes (GDEs)

KAIO Labs, HAX/SOSV

Ph.D. Research Intern

June 2025 – August 2025

New Jersey, USA

- Designed and executed CO₂ electroreduction experiments under industrially relevant conditions (>300 mA cm⁻²)
- Spray-coated composite electrodes using metal-based catalyst inks (Au, Ni, Cu, Ag, Co)
- Systematically screened catalyst compositions (metals, solvents, ionomers) and operating conditions (temperature, pressure, flow rate, electrolyte/ionomer ratio) to map performance space
- Performed electrochemical impedance spectroscopy (EIS) and characterized electrode behavior via GC and SEM/EDS
- Achieved electrodes exceeding literature and commercial benchmarks (FE >90%) using ML-guided screening

Novo Nordisk Foundation Center for Biosustainability (DTU Biosustain)

Biological and Chemical Engineer Researcher, Intern

Feb. 2023 – July 2023

Copenhagen, Denmark

- Designed the full process of sugars extraction from macro algae for industrial feedstock use using Aspen HYSYS/Aspen Plus
- Performed a technoeconomic analysis and life cycle assessment (LCA) of the process
- Quantified and analysed the economic and environmental performances of biobased feedstocks for biotechnology using a standardized method

Norwegian Institute of Bioeconomy Research (NIBIO)

Wood Technology Researcher, Intern

May 2022 – June 2022

As, Norway

- Valorized treated wood waste through fabrication of particleboards for material reuse applications
- Evaluated dimensional stability (swelling), mechanical properties, and surface wettability (contact angle) of produced boards

EDUCATION

Ph.D., Chemical and Biomolecular Engineering

New York University, New York, USA

Sep. 2023 – Present

GPA: 4.0/4.0

Advisor: [Miguel A. Modestino](#)

Research Interests: CO₂ reduction, pulsed electrochemistry, catalyst activity, Bayesian Optimization & ML, experimental automation

Masters, Biological and Chemical Engineering (Erasmus Mundus Scholarship)

Technical University of Denmark, AgroParisTech (Université Paris-Saclay), Tallinn University of Technology & Université de Reims

Champagne Ardenne - Denmark, France & Estonia

Sep. 2021 – Aug. 2023

GPA: 3.95/4.0

Research Interests: Biomaterials, bioprocessing, biotechnology, circular bioeconomy, TEA, LCA

Bachelor of Engineering, Chemical Engineering

American University of Beirut, Beirut, Lebanon

Sep. 2017 – May 2021

GPA: 3.86/4.0

Courses: Numerical Methods, thermodynamics, fluid dynamics, transport phenomena, process control, reactor engineering

PROJECTS

Operando Cu K-Edge XAFS of pulsed CO₂ electroreduction in a high-current MEA: Linking second and sub-Second catalyst dynamics to product Selectivity

- Conducted high-throughput in-situ synchrotron XAS fluorescence experiments on an MEA cell to resolve catalyst oxidation state and coordination environment under pulsing electrochemistry
- Fabricated a novel MEA cell compatible with operando X-ray measurements operating at current densities >300 mA cm⁻² and C2+ selectivity >50%

Automated optimization of dilute CO₂ electrolyzers using pulsing and active learning algorithms

- Implemented electrochemical pulsing protocols for dilute industrial CO₂ streams, improving product selectivity by 207% and energy productivity by 241% relative to steady-state operation
- Built an automated MEA CO₂ electrolyzer platform integrating LabVIEW, MATLAB, and ML algorithms to identify optimal conditions across varying feed compositions, with real-time adaptation to process disturbances
- Developed adaptive learning algorithms for rapid optimization of Cu-based multiproduct CO₂ electrolyzers under dynamic, renewable-relevant power profiles, maintaining <20% product variability across 20–100% power fluctuations

Lignin engineering for the production of a multifunctional additive for plastics

- Chemically modified Organosolv lignin via chloromethylation and esterification, confirming functionalization by FTIR with up to 92.5% mass recovery
- Fabricated PLA/modified lignin composites, showing improved processing instabilities and thermal degradation behavior
- Characterized pine and barley straw lignin via SEC, FTIR, and 2D-HSQC NMR, identifying both as superior feedstocks to aspen

L'Oréal Brandstorm Competition 2020: Building a plastic-less future in the beauty industry (Reached Top 3 at country level)

- Pioneered a sustainable bioplastic packaging solution from coffee waste for L'Oréal, validated through LCA
- Performed techno-economic and market analysis to assess production feasibility and consumer demand

PUBLICATIONS

Pulsed Electrolysis Promotes Catalyst Activity in Dilute CO₂ Streams

[ACS Energy Letters, 2026](#)

Ranking economic and environmental performance of feedstocks used in bio-based production systems

[Current Research in Biotechnology, 2025](#)

Metabolic Engineering Conference, 2024

Reimagining Unit Operations Labs: NYU's Multi-Tiered Experiential Learning Platform.

[AIChE Annual Meeting, 2025](#)

PATENTS

METHODS OF PULSED ELECTROLYSIS FOR REDUCTION OF CARBON DIOXIDE FROM MIXED GAS STREAMS

Application No. 63/996,089 and approved filing receipt by USPTO on April 22, 2026

AWARDS, FELLOWSHIPS, AND CERTIFICATIONS

Advanced Light Source Doctoral Fellowship

Granted by Lawrence Berkeley National Lab 2026 - UC Berkeley

School of Engineering (SoE) Fellowship

Granted by New York University for a Ph.D. in Chemical Engineering for the year of 2023-2024

Postgraduate program in Data Science and Business Analytics (Online)

Granted by Purdue University. Program from June 2022 – Jan 2023, Certificate ID: 65736575

SKILLS

- GC, MS, NMR, FTIR, UV-Vis, X-ray spectroscopy, SEM/EDS, spray coating, sputtering, CO₂ reduction, chromatography, PCR, gel electrophoresis, Western blot, RNA/DNA/lipids extraction, cell culture, genetic engineering
- LabVIEW, Python, Matlab, R, COMSOL, Simulink, ML-based optimization and automation, AutoCAD, LCA, TEA, Aspen HYSYS

MISC.

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Personal Site:

Available for internship: June 2027